

Wind Farm Layout Optimization

Using Aeroacoustic Surrogate Modeling and OpenFAST Validation

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Abstract

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1 Introduction

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1.1 Motivation

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1.2 Research Objective

The main objective of this study is to develop and evaluate a computational framework for wind-farm layout optimization that considers not just power production, but also aeroacoustic constraints. The framework is designed to use computationally efficient models during optimization while also including a higher-fidelity OpenFAST/FAST.Farm simulation for final validation.

More specifically, this study aims to:

1. develop a machine-learning surrogate capable of approximating single-turbine OpenFAST aeroacoustics predictions;
2. integrate the aeroacoustic surrogate with FLORIS and a genetic algorithm to optimize turbine placement across multiple wind conditions;
3. account for turbine-spacing, farm-boundary, and aeroacoustic constraints during layout optimization; and
4. validate the resulting optimized layout using FAST.Farm and OpenFAST, including both farm power production and aeroacoustic performance.

2 Background

2.1 Wind Farm Layout Optimization

Wind-farm layout optimization (WFLO) concerns the placement of turbines within a specified spatial domain to improve one or more measures of farm performance while satisfying geometric, operational, and environmental constraints.

A primary consideration in WFLO is the interaction between turbines through their wakes. As a wind turbine extracts energy from the incoming flow, it produces a downstream region of reduced wind speed and increased turbulence. A downstream turbine operating within this wake therefore

experiences different inflow conditions than a turbine exposed to the undisturbed free stream, which can reduce its power production.

Wake interactions are also dependent on wind direction. Turbines that are well separated perpendicular to one inflow direction may become aligned when the wind direction changes. Similarly, changes in wind speed affect both turbine operating conditions and wake behavior. Consequently, the performance of a candidate layout generally cannot be characterized by a single wind condition and must instead be considered across a representative set or distribution of environmental conditions.

WFLO is additionally subject to spatial constraints. These may include minimum separation between turbines, restrictions imposed by the available farm area, and setbacks from property or site boundaries. Environmental constraints can introduce further limitations on otherwise favorable turbine locations. Of particular relevance to this study is aeroacoustic noise, since changing turbine locations alters the distance and direction between individual turbines and surrounding acoustic receivers.

The combination of wake interactions, changing environmental conditions, and spatial constraints produces a complex optimization problem. Evaluating large numbers of candidate layouts can become particularly computationally expensive when higher-fidelity turbine and wake simulations are required. For this reason, wind-farm layout optimization can benefit from lower-cost models and optimization techniques capable of efficiently exploring a large spatial search domain.

2.2 OpenFAST and FAST.Farm

2.3 FLORIS

2.4 NREL 5-MW Reference Wind Turbine

The NREL 5-MW Reference Wind Turbine is a hypothetical, standardized 5-MW baseline turbine developed by the National Renewable Energy Laboratory (NREL), now known as the National Laboratory of the Rockies (NLR), primarily to support offshore wind-energy research and the development and comparison of wind-turbine simulation tools.

Although originally developed for offshore system research, the reference model also includes an onshore configuration. The fundamental turbine design is retained across these configurations, while the support structure is adapted for different applications. The onshore configuration uses a tower cantilevered directly to a concrete foundation, whereas offshore configurations have been developed using support structures such as monopiles, space frames, and floating platforms.

The reference turbine is a three-bladed, upwind, variable-speed, collective-pitch machine with a rated power of 5 MW, a rotor diameter of 126 m, and a hub height of 90 m. Selected properties relevant to this study are summarized in Table 1.

The onshore configuration was selected because this study considers aeroacoustic constraints within a comparatively compact terrestrial wind farm. The optimization domain spans only 3 km, placing potential acoustic receivers at distances where turbine noise can directly influence layout selection. In contrast, utility-scale offshore wind farms may be located many kilometers from populated shoreline receptors, making community aeroacoustic constraints less directly influential on turbine placement within the farm. The onshore configuration therefore provides a more appropriate setting for investigating the tradeoff between wake-dependent power production, turbine placement, and aeroacoustic constraints.

Table 1: Selected properties of the NREL 5-MW Reference Wind Turbine.

Property	Value
Rated power	5 MW
Rotor orientation	Upwind
Number of blades	3
Rotor diameter	126 m
Hub height	90 m
Cut-in wind speed	3 m/s
Rated wind speed	11.4 m/s
Cut-out wind speed	25 m/s
Control	Variable speed, collective pitch

2.5 Aeroacoustic Modeling

2.6 Decibels and A-Weighting

2.7 Surrogate Modeling

2.8 Genetic Algorithms

3 Methodology

3.1 Overall Workflow

The optimization framework developed in this study consists of two key components: FLORIS and a custom machine-learning predictor for the steady-state aeroacoustic output of each turbine at a given receiver location. These computationally efficient approximators avoid the time and computational costs associated with repeatedly running OpenFAST, making them suitable for use within the genetic algorithm. However, they are not intended to serve as a complete replacement for a higher-fidelity simulation. For this reason, a final OpenFAST/FAST.Farm validation step is included.

During the optimization stage, a genetic algorithm evaluates candidate turbine layouts using FLORIS and the machine-learning aeroacoustic surrogate model. On the first iteration, candidates, also referred to as chromosomes, are generated from a uniform random distribution within the constraints of the land plot. FLORIS is used to estimate wake interactions, effective turbine wind speeds at the hub, and overall farm power production for each candidate. The effective turbine wind speeds are then supplied to the aeroacoustic surrogate along with the candidate layouts to estimate the A-weighted sound level at each receiver per turbine per candidate.

The outputs of these models are incorporated into the genetic algorithm by providing information to calculate candidate fitness. The fitness function takes into account power production, as well as turbine-spacing, farm-boundary, and aeroacoustic constraints to output a single float metric. To evaluate candidates under a range of operating conditions, each generation is evaluated across a set of weighted wind-speed and wind-direction scenarios.

After each generation, chromosomes are selected based on fitness and used to produce the subsequent generation through crossover and random mutation. A portion of the lowest-performing population is replaced with newly generated random candidates to maintain population diversity, while the highest-performing candidate is retained through elitism. The specific genetic operators and optimization parameters are described in greater detail in the following sections.

Once the genetic algorithm reaches its termination criterion, currently defined as a fixed number of generations, the highest-performing candidate is evaluated using OpenFAST/FAST.Farm. The resulting higher-fidelity aeroacoustic predictions are then used to recalculate the candidate's acoustic performance and fitness and to evaluate the agreement between the surrogate-based optimization and the OpenFAST validation.

3.2 Aeroacoustic Surrogate Model

Repeated OpenFAST aeroacoustics simulations are computationally impractical within the genetic algorithm, where many candidate layouts must be evaluated across multiple wind conditions during each generation. A machine-learning surrogate was therefore developed to approximate the steady-state A-weighted sound level produced by an individual turbine at an arbitrary receiver location.

The surrogate was developed from single-turbine OpenFAST aeroacoustics simulations spanning a range of wind speeds and observer locations. Multiple spatial feature representations and regression algorithms were evaluated to determine the combination that most accurately reproduced the OpenFAST results. The selected model was subsequently vectorized to efficiently predict the individual turbine contributions across all turbines, receivers, and candidate layouts required during optimization.

The following sections describe the initial stages of the basic interpolator, generation of the training dataset, the feature and model selection procedure, the final surrogate implementation, and the method used to aggregate individual turbine predictions into farm-level receiver sound levels.

3.2.1 Preliminary Surrogate Designs

Before developing the machine-learning surrogate, two simpler aeroacoustic prediction approaches were investigated. The first used a simplified analytical relationship between turbine sound-power level, wind speed, and propagation distance. The second incorporated OpenFAST aeroacoustics results through interpolation over wind speed and observer azimuth. These preliminary approaches helped identify the variables and physical behavior that ultimately needed to be represented directly in the machine-learning surrogate.

Simplified Analytical Model. The initial model estimated the A-weighted sound-power level of an individual NREL 5 MW turbine as a function of incoming wind speed. A piecewise relationship was assumed in which the turbine produced no acoustic contribution below 3 m/s, increased linearly from 90 dBA to 106 dBA between 3 and 11 m/s, and remained at 106 dBA for wind speeds greater than or equal to 11 m/s, directly following from the specifications in Table 1. The assumed relationship was

$$L_w(v) = \begin{cases} 0, & v < 3, \\ 90 + 2(v - 3), & 3 \leq v < 11, \\ 106, & v \geq 11, \end{cases} \quad (1)$$

where v is the incoming wind speed in m/s and L_w is the estimated A-weighted sound-power level.

The sound-pressure level at each receiver was subsequently approximated using

$$L_p(v, d) = L_w(v) - 20 \log_{10}(d) - 8 - 0.005d, \quad (2)$$

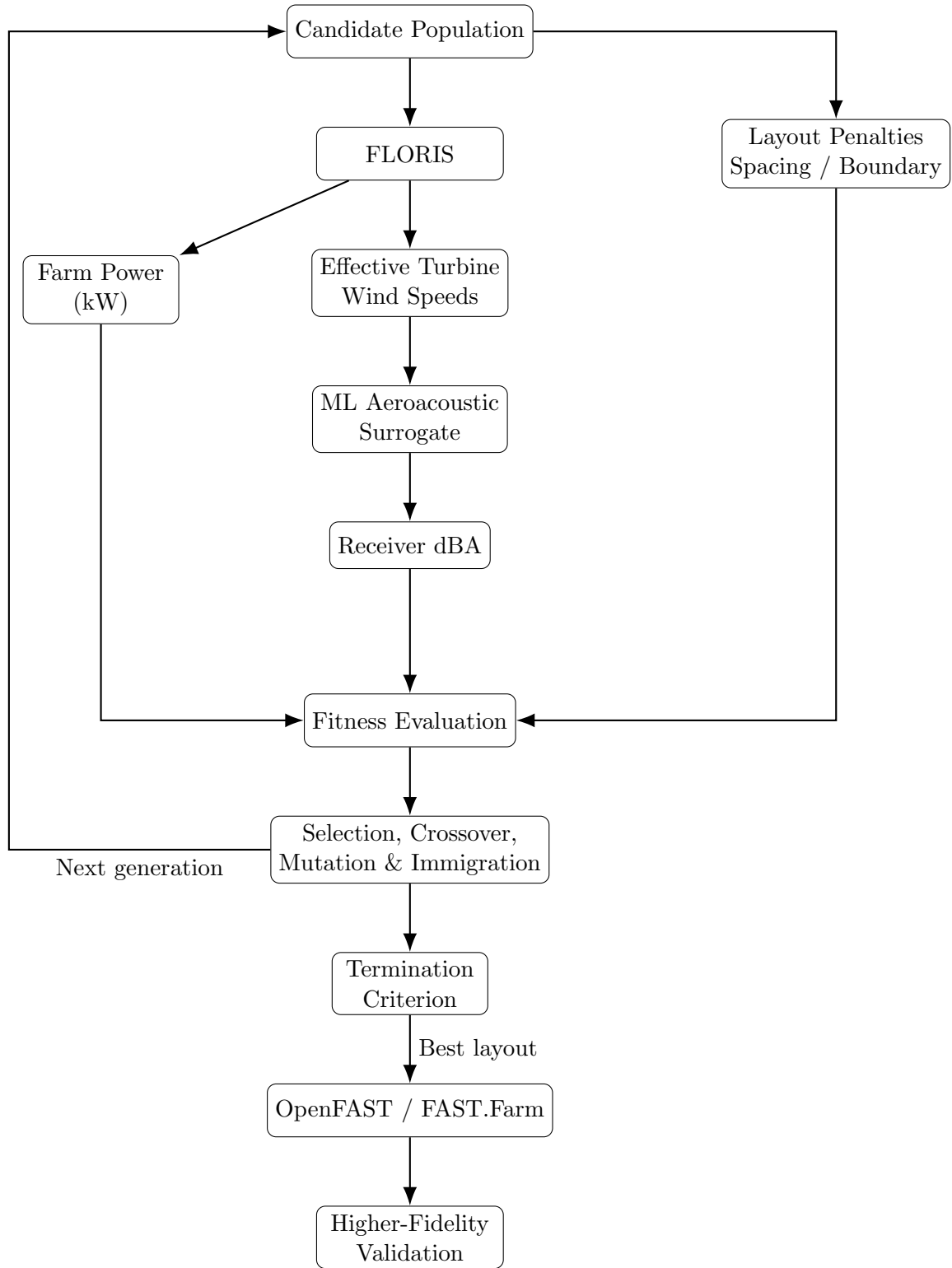


Figure 1: Overall genetic-algorithm optimization and OpenFAST/FAST.Farm validation workflow.

where d is the three-dimensional turbine-to-receiver distance. This relationship was adopted as a simplified propagation approximation during early development and should not be interpreted as a complete implementation of an established acoustic propagation standard.

Although computationally inexpensive, this formulation depends only on wind speed and turbine-to-receiver distance. It therefore predicts the same sound level for receivers at equal distances regardless of their azimuth relative to the turbine and incoming wind direction. Examination of OpenFAST aeroacoustics results showed substantial directional variation, indicating that observer azimuth needed to be represented explicitly.

OpenFAST-Based Angular Interpolator. To incorporate this directional dependence, a second surrogate was constructed from single-turbine OpenFAST aeroacoustics simulations. Wind speed was varied from 3.0 to 24.5 m/s in increments of 0.5 m/s,

$$v \in \{3.0, 3.5, 4.0, \dots, 24.5\}, \quad (3)$$

resulting in 44 distinct wind-speed conditions, following from the cut-in and cut-out speeds in Table 1. For each condition, 32 acoustic observers were uniformly distributed around the turbine at a radius of 200 m.

The latter half of each aeroacoustics output time series was retained to approximate steady-state operation. An arithmetic mean of the A-weighted sound levels over this interval was then calculated for each observer. The resulting values were stored in `noise_lookup_table.json`, forming a two-dimensional lookup table parameterized by wind speed v and observer azimuth θ .

A coordinate transformation was required when coupling this dataset with FLORIS because FLORIS specifies wind direction using the meteorological “direction from which the wind originates” convention, whereas the aeroacoustic lookup requires the corresponding Cartesian source-relative direction. The conversion used in the implementation was

$$\theta_{\text{src}} = (90^\circ - \phi_{\text{met}}) \frac{\pi}{180^\circ}, \quad (4)$$

where ϕ_{met} is the meteorological wind direction and θ_{src} is the corresponding Cartesian angle in radians.

The resulting OpenFAST data were interpolated over wind speed and azimuth using a linear `RegularGridInterpolator`. For receiver locations away from the 200 m reference radius, distance dependence was still approximated analytically using logarithmic geometric spreading and the constant 0.005 dBA/m attenuation coefficient.

This approach captured the strong directional variation absent from the initial analytical model, but the remaining analytical distance correction introduced another limitation. Comparison with additional OpenFAST simulations at varying observer distances showed that a single constant attenuation coefficient did not reproduce the distance-dependent behavior consistently across all observer azimuths and wind speeds. In some conditions the correction overestimated attenuation, while in others it underestimated it.

These observations motivated the final data-driven approach. Rather than interpolating only over wind speed and azimuth while prescribing the distance dependence analytically, the subsequent surrogate was trained to learn the joint relationship between wind speed, observer distance, observer direction, and A-weighted sound level directly from OpenFAST aeroacoustics data. A larger, homogeneous dataset was therefore generated for machine-learning model development, as described in the following sections.

It should be noted that this preliminary implementation averaged decibel values directly in the logarithmic domain. The later OpenFAST validation methodology instead uses energetic averaging, as described in Section 3.2.6.

3.2.2 Aeroacoustic Training Data

The training data for the aeroacoustic surrogate were generated using single-turbine OpenFAST simulations. Because each turbine was simulated independently, wake interactions between turbines were not included in the training-data generation process. Wake effects are instead accounted for later in the optimization workflow using FLORIS, which provides the effective wind speed experienced by each turbine.

Two primary variables were varied when generating the aeroacoustic training data: turbine wind speed and the position of the acoustic observer relative to the turbine. The wind-speed range was the same as Set (3).

Observer locations were defined in polar coordinates using radial distance r and azimuth angle θ . Twenty radial distances were generated using logarithmic spacing between 100 and 1000 m. The reference radius of 200 m was additionally included from the previous surrogate iterations to provide a higher-resolution angular sampling at a representative observer distance. Each of the logarithmically spaced radii was evaluated at 16 azimuth angles, while the 200 m reference radius was already evaluated at 32 azimuth angles.

The resulting observer sampling therefore contained

$$(20 \times 16) + 32 = 352 \quad (5)$$

observer positions. Combining these positions with the 44 wind-speed conditions produced a total of

$$N_{\text{samples}} = 44 \times 352 = 15,488 \quad (6)$$

single-turbine aeroacoustic training samples.

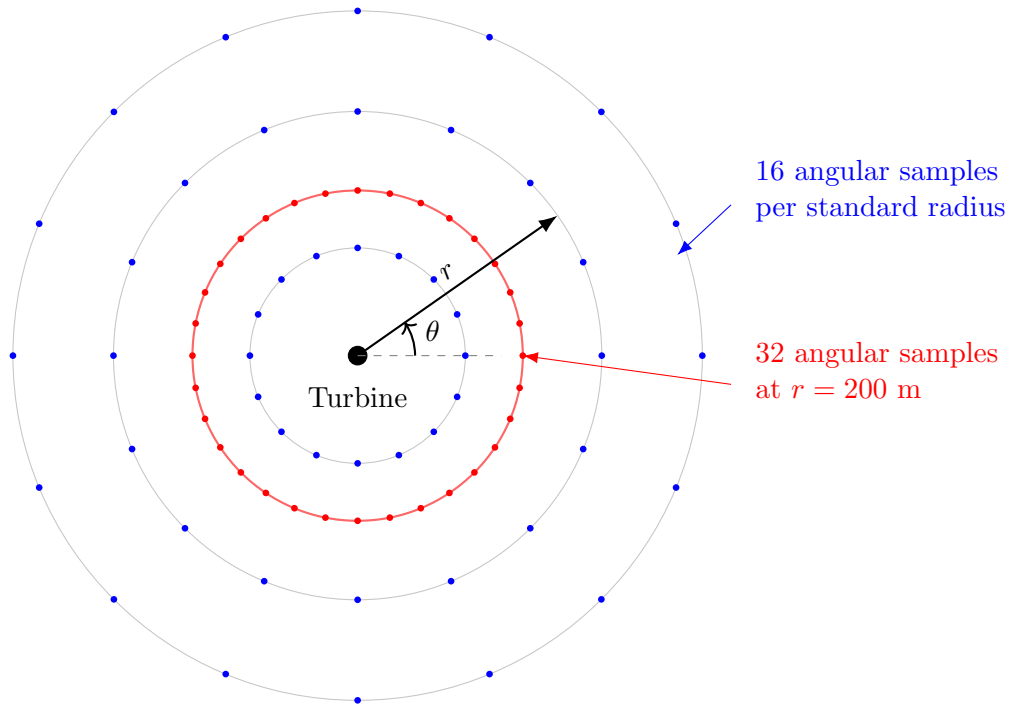
Table 2 summarizes the sampling strategy.

Table 2: OpenFAST aeroacoustic training-data sampling.

Variable	Sampling	Number of Values
Wind speed v	3.0–24.5 m/s, 0.5 m/s increments	44
Radial distance r	Logarithmically spaced, 100–1000 m	20
Azimuth θ	16 angles per logarithmically spaced radius	16
Reference radius	200 m with 32 azimuth angles	32 observer positions
Total observer positions	$(20 \times 16) + 32$	352
Total training samples	44×352	15,488

3.2.3 Candidate Feature Representations

The data generated from the OpenFAST simulations initially contained three input variables: wind speed v , relative observer coordinate X , and relative observer coordinate Y . Although the Cartesian coordinates fully describe the observer location, the aeroacoustic field exhibits a strong radial and



Schematic only: radial spacing is logarithmic from 100–1000 m

Figure 2: Schematic representation of the observer sampling used to generate the single-turbine aeroacoustic training dataset. Standard radial distances use 16 uniformly distributed azimuth angles, while the 200 m reference radius uses a denser 32-angle sampling.

directional structure. Several alternative feature representations were therefore evaluated to make this underlying geometry more explicit to the machine-learning models.

The relative Cartesian coordinates were first transformed into polar coordinates according to

$$r = \sqrt{X^2 + Y^2}, \quad (7)$$

and

$$\theta = \text{atan2}(Y, X). \quad (8)$$

Directly using θ as a numerical feature introduces an artificial discontinuity at the angular boundary. For example, directions immediately below 2π and immediately above 0 are physically adjacent but numerically distant. To preserve the periodic nature of the observer azimuth, the angular coordinate was therefore encoded using

$$c_\theta = \cos(\theta), \quad s_\theta = \sin(\theta). \quad (9)$$

Distance was also represented using several transformations. Under geometric spreading, acoustic intensity decreases with increasing distance, while sound-pressure level is expressed on a logarithmic scale. A logarithmic transformation of the observer distance,

$$r_{\log} = \log_{10}(r), \quad (10)$$

was therefore evaluated as a candidate feature. Reciprocal distance terms were additionally considered,

$$r^{-1} = \frac{1}{r}, \quad r^{-2} = \frac{1}{r^2}, \quad (11)$$

to determine whether explicitly representing distance-decay relationships improved predictive performance.

Rather than selecting one representation a priori, five candidate feature sets were constructed and evaluated with each candidate machine-learning model. These feature sets are summarized in Table 3. The complete set of original and engineered features is retained in `training_data.csv`.

Table 3: Candidate feature representations evaluated for the aeroacoustic surrogate.

Feature Set	Input Features	Purpose
Cartesian	v, X, Y	Original spatial representation
Polar	$v, r, \cos \theta, \sin \theta$	Explicit radial and directional representation
Polar + log distance	$v, \log_{10}(r), \cos \theta, \sin \theta$	Logarithmic distance representation
Polar + raw/log distance	$v, r, \log_{10}(r), \cos \theta, \sin \theta$	Combined raw and logarithmic distance
Polar + distance decay	$v, \log_{10}(r), r^{-1}, r^{-2}, \cos \theta, \sin \theta$	Multiple representations of distance decay

The feature representation was not selected independently of the regression model. Instead, each candidate machine-learning algorithm was trained and evaluated using all five feature sets. The final combination of model architecture and feature representation was selected based on predictive performance, as described in the following section.

3.2.4 Model Training and Selection

Seven regression algorithms were evaluated for the aeroacoustic surrogate: Linear Regression, Ridge Regression, Polynomial Ridge Regression, Random Forest, Extra Trees, Histogram Gradient Boosting, and eXtreme Gradient Boosting (XGBoost). These models were selected to provide a range of model complexities, from simple linear baselines to nonlinear ensemble and gradient-boosting methods. The normal Gradient Boosting was not considered due to the amount of rows in the training data. For $\sim 16k$ rows, XGB and HistGrad would be much more performant.

On that same note, standard Kriging and Gaussian Process Regression (GPR) would also run out of memory. The computational costs of $\mathcal{O}(N^3)$ time and $\mathcal{O}(N^2)$ space complexity would not outweigh the additional benefits of having a complete predictive probability distribution rather than a single point estimate. For around 16k rows, this would require an estimated 4 billion operations and 256 million matrix elements, which would take up ~ 2 GB of RAM for just one matrix.

Table 4 summarizes the candidate regression algorithms.

Table 4: Candidate regression algorithms evaluated for the aeroacoustic surrogate.

Model	Model Type	Role in Comparison
Linear Regression	Linear	Simple baseline model.
Ridge Regression	Regularized linear	Linear model with L_2 regularization applied.
Polynomial Ridge	Nonlinear polynomial	Higher-degree polynomial model with L_2 to avoid overfitting.
Random Forest	Tree ensemble	Nonlinear ensemble based on averaging predictions from multiple decision trees.
Extra Trees	Tree ensemble	Highly randomized tree ensemble providing an alternative to Random Forest.
Histogram Gradient Boosting	Gradient boosting	Nonlinear boosted-tree model designed for efficient regression on continuous data.
XGBoost	Gradient boosting	Alternative gradient-boosted tree implementation used to compare performance with Histogram Gradient Boosting.

Model architecture and feature representation were evaluated jointly. Each of the seven regression algorithms was trained using each of the five feature representations described in Table 3, resulting in

$$N_{\text{configurations}} = 7 \times 5 = 35 \quad (12)$$

candidate model configurations.

The dataset was divided into training and testing subsets using an 80/20 split. The same split was used for all model and feature-set combinations to provide a consistent basis for comparison.

Predictive performance was evaluated primarily using mean absolute error (MAE) and root mean squared error (RMSE), between the A-weighted sound level obtained from OpenFAST aeroacoustics, L_i , and the corresponding surrogate prediction $\hat{L}_i$. The coefficient of determination, R^2 , was also calculated as an additional measure of model fit.

The final surrogate configuration was selected based on its performance on the held-out test data. The comparative results of the 35 candidate configurations are presented in Section 4.1.

3.2.5 Final Surrogate Implementation

From our results, we can see that HistGrad with the `polar_log10_r` feature set performed the best, with the lowest MAE, RMSE, and the highest R^2 of all 35 model configurations. Thus, the machine-learning surrogate uses the input feature vector

$$\mathbf{x} = [v, \log_{10}(r), \cos(\theta), \sin(\theta)]. \quad (13)$$

The surrogate then approximates the relationship

$$\hat{L}_A = f(v, \log_{10}(r), \cos(\theta), \sin(\theta)), \quad (14)$$

where $\hat{L}_A$ is the predicted A-weighted sound level.

3.2.6 Aeroacoustic Aggregation

As mentioned in Section 2.6, sound is measured on a logarithmic scale, not a linear scale. Thus, all operations on sets of dB readings must be done after converting to a power intermediary, and the output can be converted back to dB. For this reason, independent turbine noise contributions are combined in the linear acoustic-energy domain as such:

$$L_{\text{total}} = 10 \log_{10} \left(\sum_{i=1}^N 10^{L_i/10} \right), \quad (15)$$

modelling the energetic equivalent of the sum operation.

Time-varying sound levels may similarly be reduced using

$$L_{\text{eq}} = 10 \log_{10} \left(\frac{1}{T} \sum_{t=1}^T 10^{L_t/10} \right), \quad (16)$$

modelling the energetic equivalent of the mean operation.

3.3 Genetic Algorithm

A genetic algorithm was implemented to optimize the layout of the turbines within the defined farm boundaries. Each candidate was evaluated through FLORIS and the machine-learning surrogate described above. Their outputs were combined with turbine-spacing and boundary constraints to produce a single scalar fitness value used to rank chromosomes.

The optimization was performed over a fixed-size population for a specified number of generations. At each generation, candidate layouts were evaluated across the prescribed weighted wind scenarios before selection, crossover, mutation, immigration, and elitism were applied to construct the subsequent generation. The following sections describe the representation of candidate layouts, fitness calculation, environmental scenarios, and genetic operators used in the optimization.

3.3.1 Chromosome Representation and Initialization

Each chromosome represents a complete candidate wind-farm layout and is stored as an $N \times 2$ matrix, where N is the number of turbines. Each row corresponds to one turbine, while the two columns contain its Cartesian x and y coordinates:

$$\mathbf{C} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_N & y_N \end{bmatrix} \in \mathbb{R}^{N \times 2}. \quad (17)$$

The initial population is generated randomly within the rectangular farm boundary. For each turbine, the x and y coordinates are independently sampled from uniform distributions,

$$x_i \sim \mathcal{U}(X_{\min}, X_{\max}), \quad y_i \sim \mathcal{U}(Y_{\min}, Y_{\max}), \quad i = 1, \dots, N. \quad (18)$$

No minimum turbine-spacing constraint is enforced during initialization. Consequently, the initial population may contain layouts in which turbines are positioned closer together than the prescribed minimum spacing. Rather than rejecting or repairing these chromosomes, spacing violations are handled through the fitness function, which penalizes layouts in proportion to the magnitude of the violation. This allows the genetic algorithm to explore the full spatial search domain while progressively favoring layouts that satisfy the spacing requirement.

For this same reason, FLORIS may sometimes warn that two hubs may be too close to each other. This is simply a by-product of the complete random generations, but does not cause any side effects besides that specific chromosome receiving a massive hit in terms of fitness, both from power lost in wake effects and the explicit C_{boundary} penalty mentioned below.

3.3.2 Wind Scenarios

Wind-farm performance depends strongly on both wind speed and wind direction. Evaluating a candidate layout under a single wind condition could therefore favor a layout that performs well under that particular condition but poorly under other likely operating conditions. To account for this variability, each chromosome is evaluated across a discrete set of weighted wind scenarios.

Wind speed and wind direction are each represented by a set of discrete values with corresponding probability weights. The Cartesian product of these distributions forms the complete set of wind scenarios. A scenario s is represented as

$$\mathcal{S}_s = (v_s, \phi_s, w_s), \quad (19)$$

where v_s is the free-stream wind speed, ϕ_s is the meteorological wind direction, and w_s is the probability weight associated with the scenario.

If wind speed v_i has weight $w_i^{(v)}$ and wind direction ϕ_j has weight $w_j^{(\phi)}$, the joint scenario weight is calculated as

$$w_{ij} = w_i^{(v)} w_j^{(\phi)}. \quad (20)$$

Assuming the individual wind-speed and wind-direction distributions are normalized,

$$\sum_i w_i^{(v)} = 1, \quad \sum_j w_j^{(\phi)} = 1, \quad (21)$$

the resulting scenario weights also satisfy

$$\sum_{s=1}^S w_s = 1, \quad (22)$$

where S is the total number of wind scenarios.

Table 5 shows the example scenarios used in testing.

Table 5: Wind-speed and wind-direction distributions used during optimization.

Variable	Value	Weight
Wind speed	11.0 m/s	0.10
	11.5 m/s	0.80
	12.0 m/s	0.10
Wind direction	260.0°	0.20
	270.0°	0.60
	280.0°	0.20

3.3.3 Fitness Function

Each chromosome is assigned a scalar fitness value that balances expected farm power production against turbine-spacing, farm-boundary, and aeroacoustic constraints. The optimization objective is expressed as

$$F = P_{\text{expected}} - C_{\text{spacing}} - C_{\text{noise}} - C_{\text{boundary}}, \quad (23)$$

where P_{expected} is the probability-weighted farm power and the remaining terms are penalties applied to constraint violations.

The individual terms are not reduced in the same manner. Power and aeroacoustic behavior depend on the wind scenario and must therefore be evaluated across the complete wind distribution. In contrast, turbine spacing and farm-boundary violations depend only on the chromosome geometry and are consequently calculated once per layout.

Expected Farm Power. As defined in Section 3.3.2, each chromosome is evaluated under S wind scenarios with normalized weights w_s .

For a farm containing N turbines, FLORIS first provides the steady-state power $P_{i,s}$ of turbine i under scenario s . The total farm power for that scenario is

$$P_s = \sum_{i=1}^N P_{i,s}. \quad (24)$$

The expected farm power used by the fitness function is then the probability-weighted average across all scenarios,

$$P_{\text{expected}} = \sum_{s=1}^S w_s P_s. \quad (25)$$

Turbine-Spacing Penalty. Turbine spacing is evaluated directly from the chromosome geometry and is therefore independent of wind scenario. For turbines i and j , their planar separation is

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}. \quad (26)$$

The default minimum allowable spacing is three rotor diameters,

$$d_{\min} = 3D, \quad (27)$$

where D is the turbine rotor diameter.

Only separations below this threshold are penalized. The total spacing penalty is therefore

$$C_{\text{spacing}} = k_{\text{spacing}} \sum_{i < j} \max(d_{\min} - d_{ij}, 0), \quad (28)$$

where k_{spacing} is a scaling coefficient that controls the fitness cost of each meter of spacing violation. Each turbine pair is included only once in the summation.

Farm-Boundary Penalty. A setback distance of

$$d_{\text{setback}} = 200 \text{ m} \quad (29)$$

is maintained between each turbine and the rectangular farm boundary.

Since the plot is implemented as a simple rectangle, the distances to the four boundaries for turbine i are easily represented by

$$\mathbf{b}_i = \begin{bmatrix} x_i - X_{\min} \\ X_{\max} - x_i \\ y_i - Y_{\min} \\ Y_{\max} - y_i \end{bmatrix}. \quad (30)$$

A soft penalty is applied whenever any of these distances falls below the required setback:

$$C_{\text{setback}} = k_{\text{setback}} \sum_{i=1}^N \sum_{e=1}^4 \max(d_{\text{setback}} - b_{i,e}, 0). \quad (31)$$

If a turbine lies completely outside the farm boundary, at least one $b_{i,e}$ becomes negative. These cases receive an additional large penalty consisting of a fixed term and a component proportional to the distance outside the boundary,

$$C_{\text{OOB}} = \begin{cases} 0, & \text{if all } b_{i,e} \geq 0, \\ k_{\text{flat}} + k_{\text{OOB}} \sum_{b_{i,e} < 0} |b_{i,e}|, & \text{otherwise.} \end{cases} \quad (32)$$

The complete boundary penalty is consequently

$$C_{\text{boundary}} = C_{\text{setback}} + C_{\text{OOB}}. \quad (33)$$

Aeroacoustic Penalty. Aeroacoustic behavior is scenario dependent. For every wind scenario, the surrogate predicts the total A-weighted sound level at each of the M boundary receivers. The controlling acoustic value for scenario s is the loudest receiver,

$$L_{\max,s} = \max_{m=1,\dots,M} L_{m,s}. \quad (34)$$

Given an allowable sound-level limit L_{limit} , only values above the limit contribute to the penalty:

$$C_{\text{noise},s} = k_{\text{noise}} \max(L_{\max,s} - L_{\text{limit}}, 0). \quad (35)$$

Two scenario-reduction methods are supported. In the default weighted mode, the final acoustic penalty is the probability-weighted average of the scenario-specific penalties,

$$C_{\text{noise}}^{\text{weighted}} = \sum_{s=1}^S w_s C_{\text{noise},s}. \quad (36)$$

Alternatively, a conservative worst-case mode may be used,

$$C_{\text{noise}}^{\text{worst}} = \max_{s=1,\dots,S} C_{\text{noise},s}. \quad (37)$$

The selected value of C_{noise} is then substituted into Equation 23.

Table 6: Reduction methods used by the fitness-function components.

Component	Within-Scenario Reduction	Across-Scenario Reduction
Farm power	Sum turbine powers	Probability-weighted average
Aeroacoustics	Energetically aggregate turbines, then take maximum across receivers	Weighted average of violation penalties or worst-case penalty
Turbine spacing	Sum pairwise spacing violations	None; geometry is scenario independent
Farm boundary	Sum setback and out-of-bounds violations	None; geometry is scenario independent

3.3.4 Selection and Genetic Operators

A fixed population size of 100 chromosomes is maintained throughout the optimization. After all chromosomes in a generation have been evaluated, the subsequent generation is constructed using a combination of elitism, tournament selection, crossover, mutation, and random immigration. These operators are intended to preserve high-performing solutions while maintaining sufficient population diversity to continue exploring the search space.

Population The default population size is

$$N_{\text{pop}} = 100. \quad (38)$$

The population size remains constant between generations. Each new generation consists of one preserved elite chromosome, offspring generated through selection and crossover, and newly initialized random immigrants.

Elitism The highest-fitness chromosome found during the optimization is preserved without modification and inserted directly into the subsequent generation. The current implementation therefore uses an elite count of one,

$$N_{\text{elite}} = 1. \quad (39)$$

This guarantees that the best solution discovered by the genetic algorithm cannot be lost through crossover, mutation, or random replacement.

Random Immigration To maintain population diversity, a fixed proportion of each new generation is reserved for independently generated random chromosomes. The number of immigrants is calculated from the population size and immigration rate according to

$$N_{\text{immigration}} = \lfloor r_{\text{immigration}} N_{\text{pop}} \rfloor, \quad (40)$$

where $r_{\text{immigration}}$ is the configured immigration rate. With the default values

$$N_{\text{pop}} = 100, \quad r_{\text{immigration}} = 0.40, \quad (41)$$

the algorithm introduces

$$N_{\text{immigration}} = 40 \quad (42)$$

new random chromosomes per generation.

These chromosomes are generated using the same uniform initialization procedure defined in Equation 18. The remaining population capacity, excluding the preserved elite chromosome, is filled with offspring produced through selection, crossover, and possible mutation.

Random immigration continually introduces unexplored layouts into the population, helping preserve genetic diversity and reducing the likelihood of premature convergence around a locally favorable region of the search space.

The complete population of the subsequent generation can therefore be expressed as

$$N_{\text{pop}} = N_{\text{elite}} + N_{\text{offspring}} + N_{\text{immigration}}. \quad (43)$$

Using the default parameters, the 100 chromosomes in each new generation consist of one preserved elite chromosome, 59 offspring generated through selection and crossover, and 40 randomly initialized immigrants:

$$100 = 1 + 59 + 40. \quad (44)$$

Parent Selection and Crossover The remaining positions in the new generation are populated by offspring. Parents are selected independently using tournament selection. For each parent, four chromosomes are sampled randomly from the current population,

$$N_{\text{tournament}} = 4, \quad (45)$$

and the chromosome with the highest fitness among those four is selected as the parent. This procedure is repeated independently to select a second parent.

Crossover is then performed at the individual coordinate level. Given two parent chromosomes $\mathbf{C}^{(1)}$ and $\mathbf{C}^{(2)}$, a binary mask $\mathbf{M}$ of the same $N \times 2$ dimensions is generated, with each element independently sampled as

$$M_{ij} \sim \text{Bernoulli}(0.5). \quad (46)$$

The resulting child chromosome is

$$C_{ij}^{(\text{child})} = \begin{cases} C_{ij}^{(1)}, & M_{ij} = 1, \\ C_{ij}^{(2)}, & M_{ij} = 0. \end{cases} \quad (47)$$

Consequently, each x or y coordinate has an equal probability of being inherited from either parent.

Mutation Newly generated offspring are additionally subject to mutation with a default probability of

$$p_{\text{mutation}} = 0.35. \quad (48)$$

When an offspring chromosome is selected for mutation, a normally distributed displacement is independently added to each of its coordinate values. For chromosome $\mathbf{C}$, mutation can be expressed as

$$\mathbf{C}' = \mathbf{C} + \boldsymbol{\epsilon}, \quad (49)$$

where

$$\epsilon_{ij} \sim \mathcal{N}(0, \sigma_{\text{mutation}}^2). \quad (50)$$

The default mutation spread is

$$\sigma_{\text{mutation}} = 100 \text{ m}. \quad (51)$$

Thus, mutation produces unbiased random spatial perturbations with a standard deviation of 100 m in both coordinate dimensions. Mutation is applied only to offspring produced through crossover; the elite chromosome and newly generated immigrants are not mutated.

Table 7: Default genetic-algorithm parameters.

Parameter	Symbol	Default
Population size	N_{pop}	100
Elite count	N_{elite}	1
Immigration rate	$r_{\text{immigration}}$	0.40
Tournament size	$N_{\text{tournament}}$	4
Mutation probability	p_{mutation}	0.35
Mutation spread	σ_{mutation}	100 m

3.4 OpenFAST Validation

The genetic algorithm relies on FLORIS and the machine-learning aeroacoustic surrogate to make repeated layout evaluations computationally practical. These approximations allow a large number of candidate layouts and wind scenarios to be evaluated during optimization, but the final optimized

layout is additionally evaluated using FAST.Farm and OpenFAST aeroacoustics as a higher-fidelity validation step.

The highest-fitness chromosome produced by the genetic algorithm is evaluated under the same wind scenarios defined in Section 3.3.2. For each scenario, the corresponding FAST.Farm simulation is generated and executed, and the resulting aeroacoustic outputs are processed to obtain the steady-state A-weighted sound level at each receiver. These results are then used to recalculate the aeroacoustic component of the fitness function without changing the optimized turbine coordinates.

3.4.1 Wind-Direction Transformation

Rather than changing the FAST.Farm inflow direction for each wind scenario, the inflow is maintained along a fixed reference direction. The complete turbine and observer geometries are instead rotated by the corresponding wind direction before each simulation. This provides an equivalent relative orientation between the incoming wind, turbines, and acoustic observers while maintaining a consistent inflow orientation within the FAST.Farm simulations.

For a point with original coordinates (x, y) , rotation about a reference point (x_c, y_c) by an angle θ is performed using

$$\begin{bmatrix} x' - x_c \\ y' - y_c \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x - x_c \\ y - y_c \end{bmatrix}. \quad (52)$$

The same transformation is applied to both turbine and receiver coordinates, preserving their relative geometry. The rotation angle is derived from the meteorological wind direction such that each physical wind scenario is represented with the FAST.Farm inflow maintained at its reference orientation.

3.4.2 Aeroacoustic Post-Processing

For each wind scenario, the aeroacoustics output from every turbine is read and reduced to its steady-state portion. The first half of each simulation output is discarded to reduce the influence of initialization and transient behavior, leaving the latter half for acoustic validation.

Unlike the preliminary surrogate described previously, which used an arithmetic mean of the logarithmic sound levels, the validation workflow uses an energetic mean to reduce the time-varying A-weighted sound level. For a sequence of T sound levels, this is calculated using Equation 16.

Because aeroacoustics is evaluated independently for each turbine, the individual turbine contributions at a common receiver are subsequently combined in the acoustic-energy domain. For N turbines, the mean A-weighted sound levels are aggregated using Equation 15, to produce a farm-level A-weighted sound level for each receiver and wind scenario.

3.4.3 Fitness Recalculation

Following completion of the FAST.Farm simulations, the fitness of the optimized chromosome is recalculated using the higher-fidelity OpenFAST outputs. Unlike the genetic-algorithm optimization stage, in which FLORIS is used to estimate farm power and the machine-learning surrogate is used to estimate aeroacoustic levels, the validation stage obtains both farm power and aeroacoustic predictions directly from the OpenFAST simulations.

For each wind scenario s , the steady-state generator power of each turbine is extracted from the corresponding OpenFAST output file using the `GenPwr (kW)` channel. To reduce the influence of simulation initialization and transient behavior, only the latter half of the time series is retained. The steady-state power of turbine i under scenario s is calculated as

$$\bar{P}_{i,s}^{\text{OF}} = \frac{1}{T} \sum_{t=1}^T P_{i,s,t}^{\text{OF}}, \quad (53)$$

where T is the number of retained time steps in the steady-state portion of the simulation. The total OpenFAST farm power for scenario s is then

$$P_s^{\text{OF}} = \sum_{i=1}^N \bar{P}_{i,s}^{\text{OF}}, \quad (54)$$

and the expected farm power is obtained using the same scenario probabilities employed during optimization,

$$P_{\text{expected}}^{\text{OF}} = \sum_{s=1}^S w_s P_s^{\text{OF}}. \quad (55)$$

The OpenFAST aeroacoustic outputs are processed independently as described previously to obtain the maximum farm-level receiver sound level for each scenario. The corresponding scenario noise penalties are calculated using the same sound-level limit and penalty coefficients as the genetic algorithm,

$$C_{\text{noise},s}^{\text{OF}} = k_{\text{noise}} \max(L_{\text{max},s}^{\text{OF}} - L_{\text{limit}}, 0). \quad (56)$$

The scenario-specific noise penalties are then reduced using the same weighted or worst-case method selected during optimization.

The turbine-spacing and boundary penalties are not recalculated from a different model because they depend only on the geometry of the optimized chromosome. The OpenFAST-validated fitness is therefore

$$F_{\text{validated}}^{\text{OF}} = P_{\text{expected}}^{\text{OF}} - C_{\text{spacing}} - C_{\text{boundary}} - C_{\text{noise}}^{\text{OF}}. \quad (57)$$

This provides a higher-fidelity evaluation of the chromosome selected by the genetic algorithm, since both wake-dependent power production and aeroacoustic behavior are evaluated using the OpenFAST/FAST.Farm simulation framework. The resulting validation comparisons are presented in Section 4.4.

For diagnostic purposes, an additional surrogate validation basis is also calculated using the OpenFAST-derived power together with the machine-learning surrogate acoustic prediction,

$$F_{\text{surrogate,OFpower}} = P_{\text{expected}}^{\text{OF}} - C_{\text{spacing}} - C_{\text{boundary}} - C_{\text{noise}}^{\text{surrogate}}. \quad (58)$$

Because Equations 57 and 58 use the same OpenFAST-derived power and geometric penalties, their difference isolates the effect of replacing the surrogate aeroacoustic prediction with the OpenFAST aeroacoustics prediction.

4 Results

4.1 Surrogate Model Performance

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An example results table is shown in Table 8.

Table 8: Example surrogate model performance.

Model	MAE (dBA)	RMSE (dBA)	R^2
Model A	0.10	0.15	0.999
Model B	0.15	0.22	0.997
Model C	0.20	0.28	0.995

4.2 Genetic Algorithm Performance

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4.3 Optimized Layout

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4.4 OpenFAST Validation

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5 Discussion

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6 Limitations

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Example limitations could be presented as:

- OpenFAST validation currently uses deterministic uniform inflow.
- The aeroacoustic surrogate is valid only within the domain represented by its training data.
- The Genetic Algorithm does not guarantee identification of the global optimum.
- Additional turbulence conditions would require further high-fidelity simulations.

7 Future Work

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8 Conclusion

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References

- [1] Author, *Example OpenFAST Reference*, Publisher or Journal, Year.
- [2] Author, *Example FLORIS Reference*, Publisher or Journal, Year.

A Optimization Parameters

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B Additional Results

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